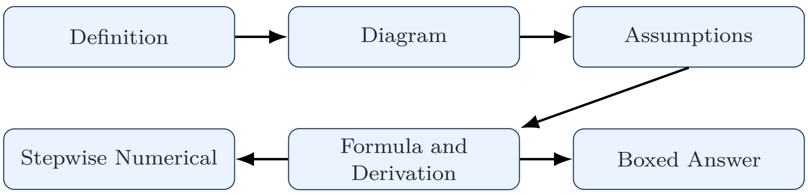
B.E. Electrical Engineering

Second Year Second Semester

Power Supply Systems

2022 Part-II Complete Solutions

Academic Article and Printed Book Style
Simple English | SN Ma'am Method | Diagrams Redrawn from Provided Class Materials



Prepared for exam practice from the 2022 Part-II question paper

All long derivations and numericals are written in a margin-safe format.

Contents

How to Read These Solutions

In Part-II, answers should not look like memorised paragraphs. The best style is:

$$\boxed{\text{Definition} \rightarrow \text{Neat diagram} \rightarrow \text{Physical meaning} \rightarrow \text{Formula} \rightarrow \text{Stepwise calculation}}$$

All diagrams below are redrawn in clean TikZ form from the provided class materials, so that labels do not overlap and the answer has a printed-book impression.

Question 1

Question asked

Compare the volume of conductor material required for a d.c. three-wire system with an a.c. two-phase three-wire system in underground cables. State the assumptions.

Given Topic

This question belongs to the standard chapter: **comparison of conductor material**. For underground cables the basis of comparison is different from overhead lines.

Assumptions used for comparison

For both systems, assume that:

1. The same power P is transmitted.
2. The same distance l is used.
3. The same power loss W is allowed in the conductors.
4. The same resistivity ρ of conductor material is used.
5. In underground cables, the maximum voltage between any two conductors is kept the same, say V_m.
6. The a.c. two-phase load has power factor $\cos\phi$.
7. In the two-phase three-wire system, the neutral area is selected for the same current density.

Diagrams from the Provided Material - Redrawn Clearly

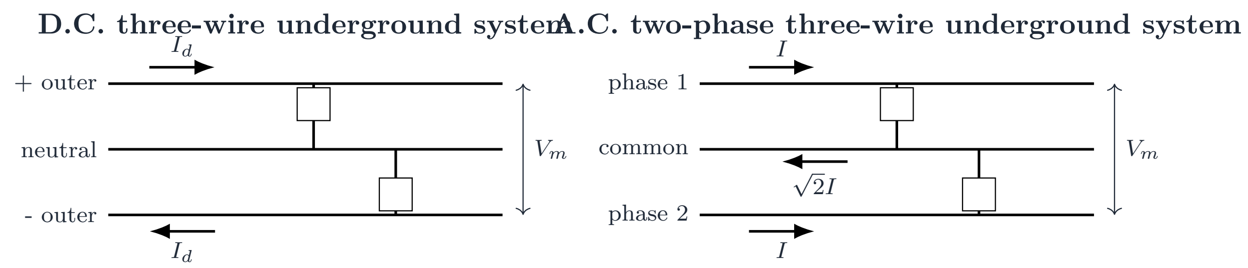


Figure 1: Underground comparison diagrams redrawn from class material.

Step 1: Basic reference quantity

Let

$$K = \frac{4P^2\rho l^2}{WV_m^2}$$

This is commonly used as the reference volume for a two-wire d.c. underground system.

Step 2: D.C. three-wire system

For balanced loading in a d.c. three-wire system,

$$I_d = \frac{P}{V_m}$$

There is no current in neutral under balanced condition. Loss occurs mainly in the two outer conductors:

$$W = 2I_d^2 \frac{\rho l}{a_d}$$

Therefore,

$$a_d = \frac{2P^2 \rho l}{WV_m^2}$$

The neutral is usually taken as half of an outer conductor. Hence volume is

$$V_{dc} = \left(2a_d + \frac{a_d}{2}\right) l = 2.5 a_d l$$

So,

$$V_{dc} = 2.5 \left(\frac{2P^2 \rho l}{WV_m^2}\right) l = \frac{5P^2 \rho l^2}{WV_m^2}$$

In terms of K,

$$\boxed{V_{dc} = \frac{5}{4}K = 1.25K}$$

Step 3: A.C. two-phase three-wire system

For a two-phase three-wire system, the voltage between the two outer wires is V_m. Hence the phase voltage is

$$V_{ph} = \frac{V_m}{\sqrt{2}}$$

Power is

$$P = 2V_{ph} I \cos\phi = 2\left(\frac{V_m}{\sqrt{2}}\right) I \cos\phi = \sqrt{2} V_m I \cos\phi$$

Therefore,

$$I = \frac{P}{\sqrt{2} V_m \cos\phi}$$

The common wire carries the phasor sum of two equal currents displaced by 90°, so

$$I_n = \sqrt{2} I$$

For the same current density,

$$a_n = \sqrt{2} a$$

Total loss is

$$W = 2I^2 \frac{\rho l}{a} + I_n^2 \frac{\rho l}{a_n} = 2I^2 \frac{\rho l}{a} + 2I^2 \frac{\rho l}{\sqrt{2} a} = (2 + \sqrt{2}) I^2 \frac{\rho l}{a}$$

So,

$$a = \frac{(2+\sqrt{2}) I^2 \rho l}{W} = \frac{(2+\sqrt{2}) P^2 \rho l}{2WV_m^2 \cos^2\phi}$$

Volume of conductor material is

$$V_{2\phi 3w} = (2a + a_n) l = (2 + \sqrt{2}) a l$$

Therefore,

$$V_{2\phi 3w} = \frac{(2+\sqrt{2})^2 P^2 \rho l^2}{2WV_m^2 \cos^2\phi}$$

In terms of K,

$$\boxed{V_{2\phi 3w} = \frac{3+2\sqrt{2}}{4\cos^2\phi}K = \frac{1.457K}{\cos^2\phi}}$$

Final Comparison

$$\frac{V_{dc}}{V_{2\phi 3w}} = \frac{1.25K}{1.457K/\cos^2\phi} = 0.858\cos^2\phi$$

Final answer: The d.c. three-wire underground system requires $1.25K$ volume, while the a.c. two-phase three-wire underground system requires $1.457K/\cos^2\phi$. Hence the d.c. three-wire system requires $0.858\cos^2\phi$ times the conductor material of the a.c. two-phase three-wire system.

Question 2

Question asked

State Kelvin's law. Applying Kelvin's law determine the most economic cross-section for the conductor of a three-phase, 33 kV, 8 km overhead line. Load cycle: 2500 kW for 8 h/day, 1500 kW for 10 h/day and 1000 kW for 6 h/day at unity power factor. The line is used for 365 days. Line cost is Rs. $(16250 + 500a)$ per km, where a is area in cm^2. Resistance per conductor per km for area $1\,\text{cm}^2$ is $0.176\,\Omega$. Energy cost is 5 paise/kWh. Interest and depreciation charges are 8%.

Statement of Kelvin's Law

Kelvin's law states that the most economical size of a conductor is obtained when:

$$\boxed{\text{Annual variable capital charge} = \text{Annual cost of energy lost}}$$

In simple words, the conductor should not be too small or too large. A small conductor gives high loss, and a very large conductor gives high capital cost.

Cost Curve Diagram

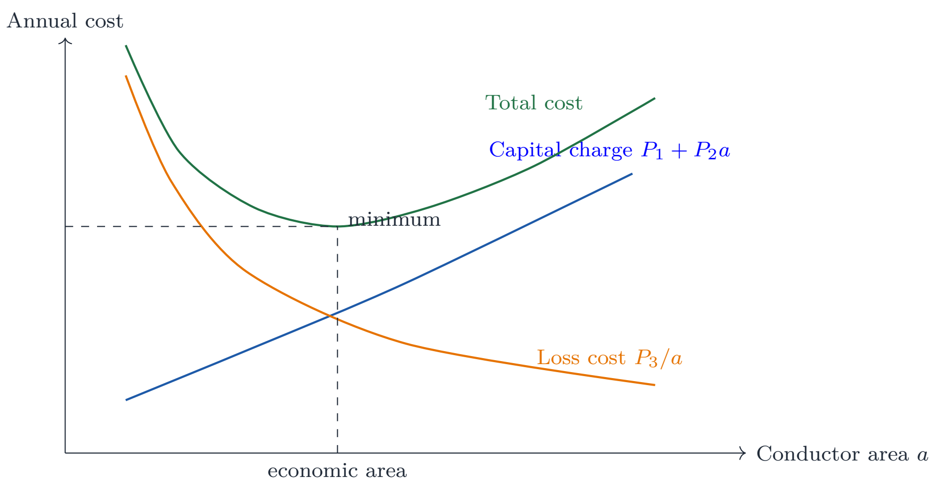


Figure 2: Kelvin's law: minimum total annual cost occurs at economical conductor area.

Given Data

Line voltage	$V_L = 33\,\text{kV}$
Length	$l = 8\,\text{km}$
Power factor	$\cos\phi = 1$
Energy cost	5 paise/kWh = Rs. 0.05/kWh
Interest and depreciation	8%
Line cost per km	Rs. $(16250 + 500a)$
Resistance per conductor per km	$0.176/a\,\Omega$

Step 1: Variable annual capital charge

Variable cost per km is

$$500a \quad \text{Rs/km}$$

For 8 km,

$$\text{variable capital cost} = 8 \times 500a = 4000a$$

Annual charge at 8% is

$$\text{annual variable charge} = 0.08 \times 4000a = 320a \quad \text{Rs/year}$$

Step 2: Currents for different loads

For a three-phase line,

$$I = \frac{P}{\sqrt{3}V_L \cos\phi}$$

Load	Duration/day	Current	Contribution I^2t
2500 kW	8 h	43.74 A	15304.56
1500 kW	10 h	26.24 A	6887.05
1000 kW	6 h	17.50 A	1836.55

So,

$$\sum I^2t = 24028.16$$

Step 3: Energy loss per year

Resistance of one conductor for 8 km is

$$R = \frac{0.176 \times 8}{a} = \frac{1.408}{a}\,\Omega$$

Daily loss energy is

$$E_d = \frac{3R\sum I^2t}{1000} = \frac{3 \times 1.408 \times 24028.16}{1000a} = \frac{101.49}{a}\,\text{kWh/day}$$

Annual energy loss is

$$E_y = 365E_d = \frac{37045.66}{a}\,\text{kWh/year}$$

Annual cost of energy lost is

$$\text{loss cost} = 0.05 \times \frac{37045.66}{a} = \frac{1852.28}{a} \quad \text{Rs/year}$$

Step 4: Apply Kelvin's Law

$$320a = \frac{1852.28}{a}$$
$$a^2 = \frac{1852.28}{320} = 5.788$$
$$a = 2.406\,\text{cm}^2$$

Final answer: The most economical cross-sectional area of each conductor is $\boxed{2.41\,\text{cm}^2}$ approximately.

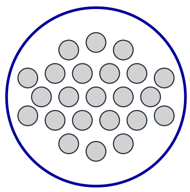
Question 3

Question asked

(i) What do you understand by "23/0.3 PVC cable"? (ii) A 50 km, 33 kV, three-phase three-wire line supplies a balanced load of 6 MW at 0.8 p.f. lagging. If transmission efficiency is 80%, determine the weight of copper required. What reduction in weight of copper can be obtained in case of single-phase two-wire a.c. line operating at the same voltage between lines? Take $\rho = 1.73 \times 10^{-8}\,\Omega\text{m}$ and density $= 8900\,\text{kg/m}^3$.

Part (i): Meaning of 23/0.3 PVC Cable

Stranded PVC Cable



23 strands, each of diameter 0.3 mm

PVC insulation is shown by outer cover

Figure 3: Meaning of cable notation 23/0.3.

Simple explanation

23/0.3 PVC cable means:

- The cable has 23 strands.
- Diameter of each strand is 0.3 mm.
- PVC means polyvinyl chloride insulation is used.

Approximate copper area is

$$A = 23 \times \frac{\pi}{4}(0.3)^2 = 1.626\,\text{mm}^2$$

Part (ii): Weight of Copper Required

Given Data

$$P_{out} = 6\,\text{MW}, \quad V_L = 33\,\text{kV}, \quad \cos\phi = 0.8, \quad \eta = 80\% = 0.8$$
$$l = 50\,\text{km} = 50000\,\text{m}, \quad \rho = 1.73 \times 10^{-8}\,\Omega\text{m}, \quad \gamma = 8900\,\text{kg/m}^3$$

Step 1: Line loss

$$\eta = \frac{P_{out}}{P_{in}} \Rightarrow P_{in} = \frac{6}{0.8} = 7.5\,\text{MW}$$

$$P_{loss} = P_{in} - P_{out} = 7.5 - 6 = 1.5\,\text{MW}$$

Step 2: Three-phase three-wire line

Line current is

$$I_3 = \frac{P}{\sqrt{3}V_L \cos\phi} = \frac{6 \times 10^6}{\sqrt{3} \times 33 \times 10^3 \times 0.8} = 131.22\,\text{A}$$

Power loss in three conductors is

$$P_{loss} = 3I_3^2 R$$

$$R = \frac{1.5 \times 10^6}{3(131.22)^2} = 29.05\,\Omega$$

Conductor area is

$$a = \frac{\rho l}{R} = \frac{1.73 \times 10^{-8} \times 50000}{29.05} = 2.978 \times 10^{-5}\,\text{m}^2$$

Total volume of copper is

$$V_c = 3al = 3 \times 2.978 \times 10^{-5} \times 50000 = 4.467\,\text{m}^3$$

Weight is

$$W_3 = V_c \gamma = 4.467 \times 8900 = 39756\,\text{kg}$$

Step 3: Single-phase two-wire line at same voltage

For single-phase,

$$I_1 = \frac{P}{V_L \cos\phi} = \frac{6 \times 10^6}{33 \times 10^3 \times 0.8} = 227.27\,\text{A}$$

$$P_{loss} = 2I_1^2 R_1$$

$$R_1 = \frac{1.5 \times 10^6}{2(227.27)^2} = 14.52\,\Omega$$

$$a_1 = \frac{\rho l}{R_1} = 5.956 \times 10^{-5}\,\text{m}^2$$

$$V_1 = 2a_1 l = 2 \times 5.956 \times 10^{-5} \times 50000 = 5.956\,\text{m}^3$$

$$W_1 = 5.956 \times 8900 = 53008\,\text{kg}$$

Reduction obtained by using three-phase line instead of single-phase line is

$$\Delta W = 53008 - 39756 = 13252\,\text{kg}$$

$$\%\text{ reduction} = \frac{13252}{53008} \times 100 = 25\%$$

Final answer: Weight of copper for the three-phase three-wire line is about $\boxed{39.8\,\text{tonnes}}$. The corresponding single-phase two-wire line would need about $\boxed{53.0\,\text{tonnes}}$. Hence three-phase transmission gives about $\boxed{25\%}$ reduction in copper weight, i.e. about $\boxed{13.25\,\text{tonnes}}$.

Question 4

Question asked

Define prospective current and pre-arcing time in a fuse. Explain feeders, distributors and service mains.

Fuse Terms

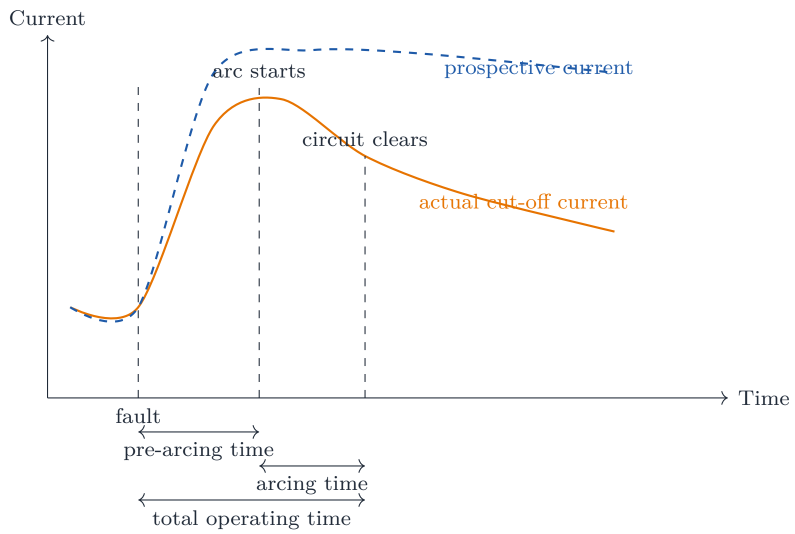


Figure 4: Fuse operating times and prospective current.

Definitions

Prospective current: It is the r.m.s. value of fault current which would flow in the circuit if the fuse were replaced by a conductor of negligible impedance. It is the possible short-circuit current of the system.

Pre-arcing time: It is the time interval from the instant of fault current start to the instant when the fuse element melts and an arc is initiated.

Feeder, Distributor and Service Mains

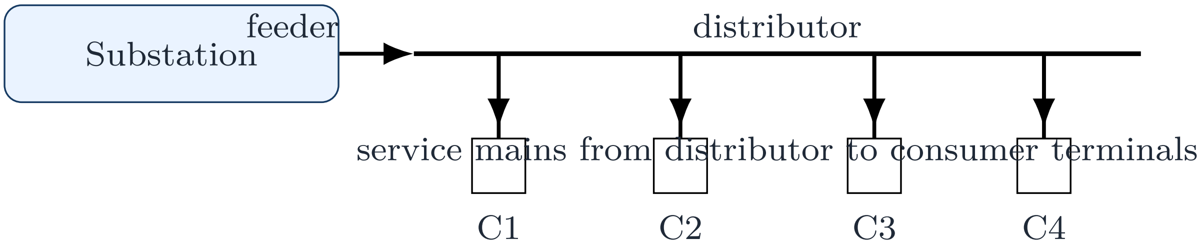


Figure 5: Feeder, distributor and service mains in a distribution system.

Item	Explanation
Feeder	A feeder connects the substation to the area to be supplied. Normally no tapping is taken from a feeder. It is designed mainly from current-carrying capacity.
Distributor	A distributor is the conductor from which several tappings are taken for consumers. It is designed mainly from the voltage-drop point of view.
Service mains	Service mains are small conductors which connect the consumer terminals to the distributor.

Exam remark: For a short answer, always write: feeder is current-capacity based; distributor is voltage-drop based. This sentence gives direct marks.

Question 5

Question asked

(i) Derive an expression for the power loss in a uniformly loaded distributor fed at both ends with equal voltages. (ii) A d.c. two-wire distributor AB, 300 m long, supplies a uniformly distributed load of 0.25 A/m. Concentrated loads of 40 A and 60 A are at P and Q respectively. AP = BQ = 120 m. The loop resistance is $0.001\,\Omega$/m. Determine currents fed at A and B and voltages at P and Q. Both A and B are maintained at 300 V.

Part (i): Power Loss in Uniformly Loaded Distributor

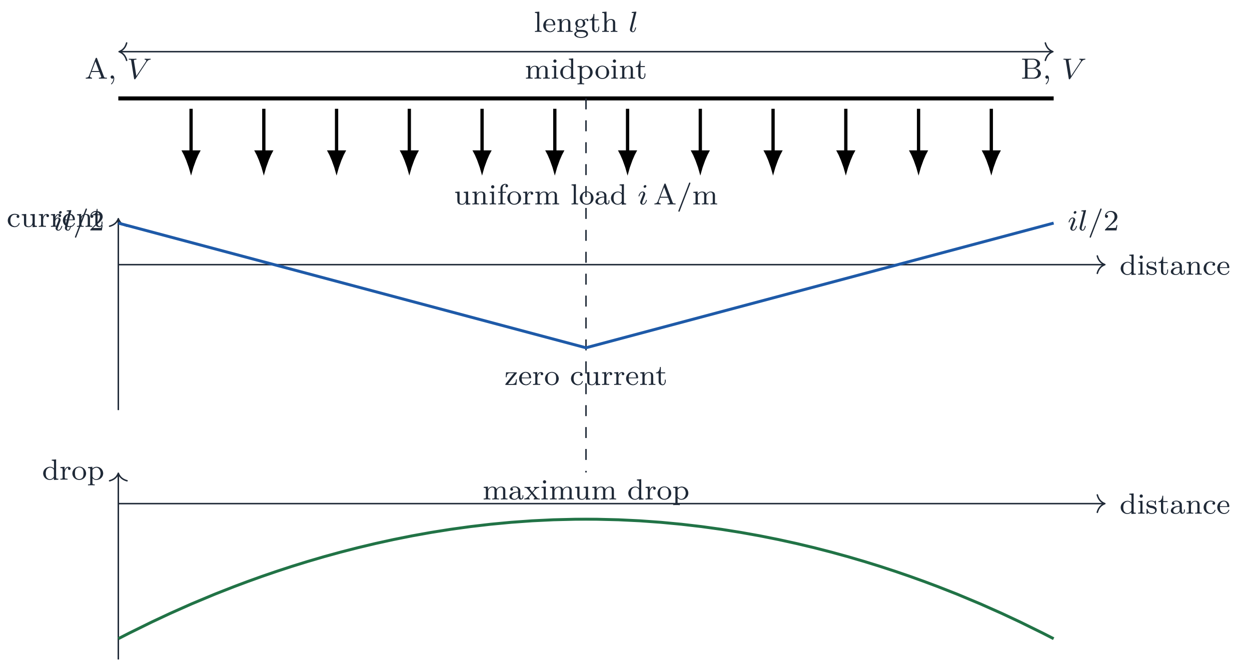


Figure 6: Uniformly loaded distributor fed at both ends with equal voltages.

Let i be the load per metre, l be the length and r be loop resistance per metre.

Since both end voltages are equal, the midpoint is the point of minimum voltage. Current supplied from each end is

$$\frac{il}{2}$$

At a distance x from the midpoint, the current in the small element is

$$ix$$

Power loss in a small length dx is

$$dP = (ix)^2 r\,dx$$

Due to symmetry, total loss in the distributor is twice the loss of half length:

$$P_{loss} = 2\int_0^{l/2} i^2x^2r\,dx$$

$$P_{loss} = 2i^2r\left[\frac{x^3}{3}\right]_0^{l/2} = 2i^2r\frac{l^3}{24}$$

$$\boxed{P_{loss} = \frac{i^2rl^3}{12}}$$

Since total current $I = il$ and total loop resistance $R = rl$,

$$\boxed{P_{loss} = \frac{I^2R}{12}}$$

Part (ii): Numerical Solution

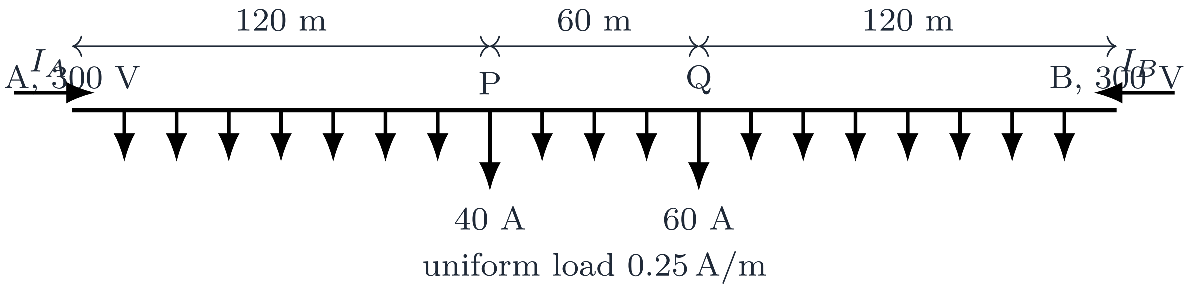


Figure 7: Given d.c. distributor for Question 5(ii).

Step 1: Total load

Uniformly distributed load is

$$0.25 \times 300 = 75\,\text{A}$$

Total concentrated load is

$$40 + 60 = 100\,\text{A}$$

Total load is

$$75 + 100 = 175\,\text{A}$$

Therefore,

$$I_A + I_B = 175$$

Step 2: Find current fed from A

Let I_A be the current fed from end A. Since end voltages are equal, total voltage drop from A to B is zero:

$$\int_0^{300} I(x) r\, dx = 0$$

Using the load positions from A,

$$I_A(300) - 0.25\frac{300^2}{2} - 40(300-120) - 60(300-180) = 0$$

$$300I_A - 11250 - 7200 - 7200 = 0$$

$$300I_A = 25650$$

$$I_A = 85.5\,\text{A}$$

So,

$$I_B = 175 - 85.5 = 89.5\,\text{A}$$

Step 3: Voltage at P

Loop resistance per metre is $r = 0.001\,\Omega/\text{m}$.

Drop from A to P:

$$\Delta V_{AP} = r\left[I_A(120) - 0.25\frac{120^2}{2}\right]$$

$$\Delta V_{AP} = 0.001\left[85.5 \times 120 - 0.25 \times \frac{14400}{2}\right]$$

$$\Delta V_{AP} = 0.001(10260 - 1800) = 8.46\,\text{V}$$

$$V_P = 300 - 8.46 = 291.54\,\mathrm{V}$$

Step 4: Voltage at Q

Drop from A to Q:

$$\Delta V_{AQ} = r\left[I_A(180) - 0.25\frac{180^2}{2} - 40(180-120)\right]$$

$$\Delta V_{AQ} = 0.001\,[15390 - 4050 - 2400] = 8.94\,\mathrm{V}$$

$$V_Q = 300 - 8.94 = 291.06\,\mathrm{V}$$

Final answer: Current supplied from A is $\boxed{85.5\,\mathrm{A}}$ and current supplied from B is $\boxed{89.5\,\mathrm{A}}$. Voltage at P is $\boxed{291.54\,\mathrm{V}}$. Voltage at Q is $\boxed{291.06\,\mathrm{V}}$.

Question 6

Question asked

(i) What do you understand by earthing? Mention the factors on which earth resistance depends. (ii) Explain the operation of protective relays in power system.

Part (i): Earthing

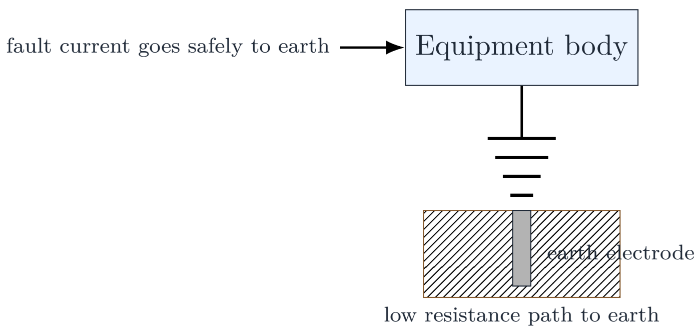


Figure 8: Basic idea of equipment earthing.

Definition

Earthing means connecting the metallic non-current carrying body of electrical equipment to the general mass of earth through a low resistance path. It protects human beings and equipment during insulation failure.

Factors Affecting Earth Resistance

Earth resistance depends on:

1. material of earth electrode and earth wire,
2. size and shape of electrode,
3. depth of burial of electrode,
4. moisture content of soil,
5. temperature of soil,
6. type and resistivity of soil,

7. use of charcoal and salt around electrode,
8. condition of joints and connections.

Part (ii): Operation of Protective Relay

A protective relay detects abnormal conditions and gives a trip signal to the circuit breaker.

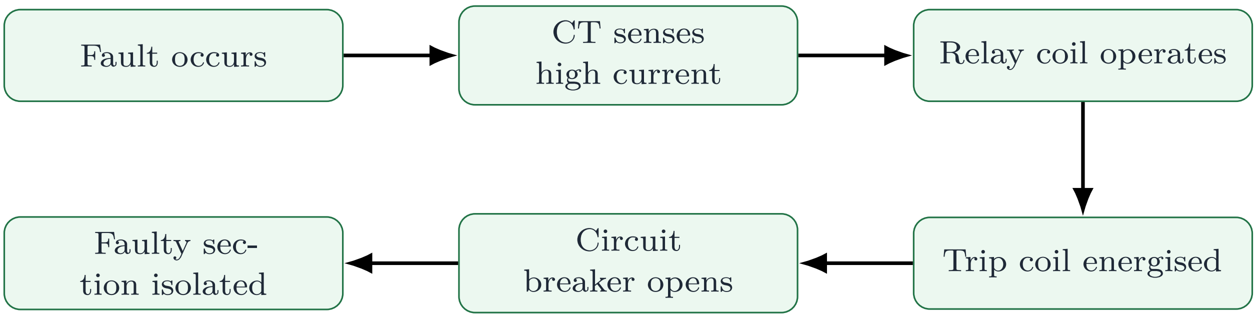


Figure 9: Operation sequence of a protective relay.

Working in simple steps

1. Under normal condition, relay contacts remain open and the circuit breaker remains closed.
2. When a fault occurs, current becomes very high.
3. The current transformer gives a proportional current to the relay coil.
4. The relay detects the fault and closes its trip contact.
5. The trip coil of the circuit breaker is energised.
6. The circuit breaker opens and isolates the faulty part.
7. Healthy parts of the system continue to operate.

Exam remark: Write relay operation as a sequence. A flowchart gives a clear and organized answer.

Question 7

Question asked

(i) What do you understand by stranded conductor? What are its advantages? (ii) Write short note on ACSR conductor.

Part (i): Stranded Conductor

Figure 10: Solid conductor and stranded conductor.

Definition

A stranded conductor is made by twisting several thin wires of small cross-sectional area together. Each small wire is called a strand. Stranding makes the conductor flexible.

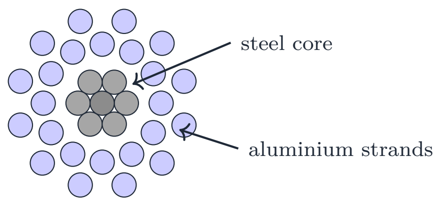
Advantages of Stranded Conductors

1. It is more flexible than a solid conductor.
2. It can be coiled and transported easily.
3. It has better mechanical strength.
4. It is suitable for long overhead transmission and distribution lines.
5. Broken strands do not immediately make the whole conductor useless.
6. It can be made with a steel core to improve strength.

Part (ii): ACSR Conductor

ACSR means **Aluminium Conductor Steel Reinforced**.

ACSR conductor cross-section



Aluminium carries current; steel gives strength.

Figure 11: ACSR conductor redrawn from provided material.

Construction and Working

- The central core is made of steel strands.
- Around the steel core, aluminium strands are placed.
- Aluminium has good conductivity and carries most of the current.
- Steel has high tensile strength and takes mechanical load.

Advantages of ACSR

1. High tensile strength.
2. Suitable for long spans.
3. Less sag compared with all-aluminium conductor.
4. Economical for overhead transmission lines.
5. Good conductivity due to aluminium.
6. Mechanical support is provided by steel core.

Final answer: ACSR is widely used in overhead transmission because it gives a good combination of conductivity and mechanical strength.

Final Revision Table

Question	Core idea	Must write in exam
Q1	Conductor comparison	Assumptions, circuit diagrams, volume ratio
Q2	Kelvin's law	Cost curve, $C = P_1 + P_2 a + P_3/a$, numerical
Q3	Cable notation and copper weight	Meaning of 23/0.3, efficiency-loss relation, 25% saving in 3-phase
Q4	Fuse and distribution terms	Prospective current, pre-arcing time, feeder-distributor-service mains diagram
Q5	Distributor fed both ends	Uniform load derivation, KVL/integral method, voltage at load points
Q6	Earthing and relay	Earthing definition, factors, relay flowchart
Q7	Stranded and ACSR	Cross-section diagrams, advantages, current and strength roles

Appendix A: SN Ma'am Answer Writing Templates

Theory answer template

Use this sequence for theory questions:

1. Write a one-line definition.
2. Draw the neat labelled diagram.
3. Explain the need of the system or device.
4. Explain construction or parts.
5. Explain working in simple steps.
6. Write advantages or limitations.
7. End with one exam remark.

Numerical answer template

Use this sequence for numerical questions:

1. Write the given data with units.
2. Draw the circuit or distributor diagram.
3. Assume current direction clearly.
4. Write section currents or current equations.
5. Apply KCL or KVL.
6. Calculate voltage drop, area, cost or loss step by step.
7. Box the final answer with unit.

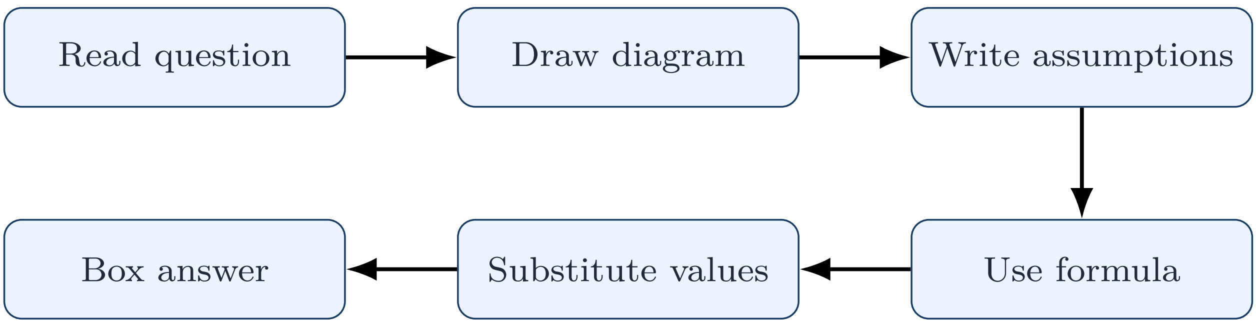


Figure 12: Safe exam-solving route for Part-II questions.

Appendix B: Important Formula Sheet

Topic	**Formula / Key result**
Kelvin's law	Variable annual capital charge = annual cost of energy lost.
Total annual cost	$C = P_1 + P_2 a + P_3/a$.
Three-phase current	$I = P/(\sqrt{3}V_L \cos\phi)$.
Single-phase current	$I = P/(V \cos\phi)$.
Line loss, three-phase	$P_{loss} = 3I^2R$.
Line loss, single-phase	$P_{loss} = 2I^2R$.
Resistance of conductor	$R = \rho l/a$.
Uniform distributor, both ends	$P_{loss} = i^2rl^3/12 = I^2R/12$.
Voltage drop section	ΔV = section current × section resistance.
Fusing factor	Fusing current / current rating.
Neutral current in d.c. 3-wire	Difference of currents on the two sides.

For distributor questions, never start with formula only. First draw the line, mark distances, then write currents in every section.

Appendix C: Diagrams to Practise from Provided Materials

No.	Diagram	Used in question
1	D.C. three-wire system	Conductor material comparison
2	A.C. two-phase three-wire system	Conductor material comparison
3	Kelvin's law cost curve	Kelvin's law proof and numerical
4	Stranded cable cross-section	Meaning of cable notation
5	Fuse time-current curve	Fuse terms
6	Feeder-distributor-service mains	Distribution definitions
7	Uniformly loaded distributor fed at both ends	Distributor derivation
8	Earthing arrangement	Earthing definition
9	Protective relay flowchart	Relay operation
10	ACSR cross-section	ACSR short note

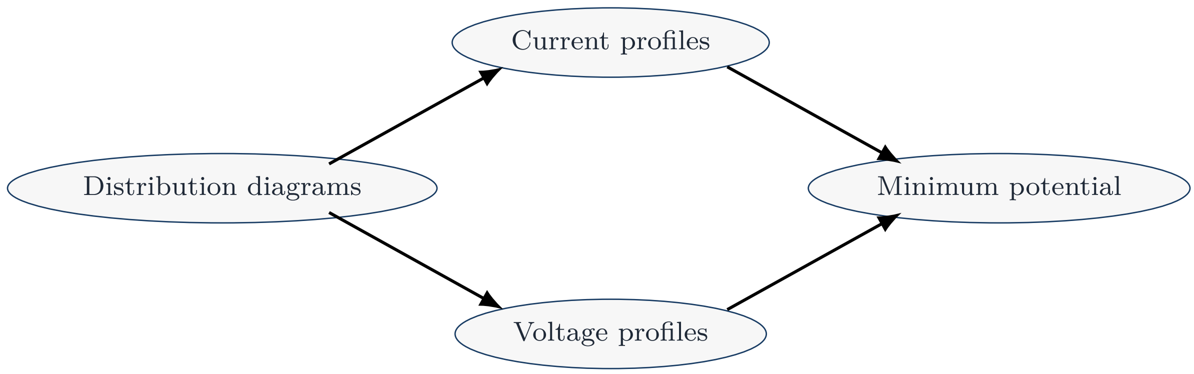


Figure 13: Idea map for distributor questions.

Appendix D: Common Mistakes to Avoid

1. Do not mix overhead and underground comparison basis. In overhead line, voltage to earth is important. In underground cable, voltage between conductors is important.
2. In Kelvin's law, do not include the fixed part of capital cost in the variable charge equation.
3. In three-phase copper-weight problems, use $3I^2R$, not $2I^2R$.
4. In a distributor fed from both ends, do not assume that one end supplies all the load.
5. In fuse questions, distinguish clearly between pre-arcing time and arcing time.
6. In feeder-distributor questions, remember: feeder has no tapping; distributor has tapping.
7. In ACSR, aluminium carries current and steel gives mechanical strength.

Final exam strategy: For Part-II, write less complicated English but draw more accurate diagrams. A clean diagram plus correct assumptions usually gives half the marks before the calculation starts.